

Integration — Lesson 11 Test: Motion I — From Velocity to Position

10 questions · show your work in the box · calculators allowed · remember: new position = starting position + $\int v \, dt$

Name: _____ Date: _____ Score: _____ / 10

1. A robot rolls at a constant velocity of 8 m/s, starting from position $s = 0$. Where is it after 12 seconds? ^{1 pt}

2. A car passes mile marker 10 driving at a steady 50 mph. What mile marker is it at 2 hours later? ^{1 pt}

3. An object's velocity is $v(t) = 2t$ m/s. Find its displacement from $t = 0$ to $t = 5$ seconds. ^{1 pt}

4. An object starts at position $s = 2$ m with velocity $v(t) = 3t^2$ m/s. Where is it at $t = 2$ seconds?

1 pt

5. A remote-control car starts at position $s = 20$ m and drives in reverse with $v = -3$ m/s for 4 seconds. Where does it end up?

1 pt

6. An object moves at $v = 5$ m/s from $t = 0$ to $t = 2$, then $v = -2$ m/s from $t = 2$ to $t = 6$. Find (a) its displacement and (b) its total distance traveled.

1 pt

7. A cart has velocity $v(t) = 8 - 2t$ m/s. (a) When does it momentarily stop? (b) How far has it traveled from $t = 0$ up to that stopping moment?

1 pt

8. An object moves with $v(t) = 4 - 2t$ m/s from $t = 0$ to $t = 4$. Find its displacement AND its total distance traveled. (Watch for the sign change!) 1 pt

9. A velocity graph, in words: $v = 6$ m/s (flat) for the first 3 seconds, then a straight line dropping from 6 down to 0 over the next 2 seconds. Using only rectangle and triangle areas, find the displacement over the whole 5 seconds. 1 pt

10. CHALLENGE: an object moves with $v(t) = t^2 - 2t$ m/s on $[0, 3]$. Find when it turns around, its net displacement, and its total distance traveled. 1 pt
